

George Spyrou

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Final-year Computer Science student at the National and Kapodistrian University of Athens, applying to the thesis-based **Master of Mathematics - Computer Science** at the University of Waterloo. My research so far has been in machine learning and knowledge graph question answering, and I plan to continue into a PhD with the goal of a career in research and teaching.

Education

National and Kapodistrian University of Athens (NKUA)

2022 – 2026 (expected)

BSc, Department of Informatics and Telecommunications. Specialization: Data and Knowledge Management

- Cumulative GPA: 7.71/10, with a **Years 3–4 GPA of 7.93/10**. My grades rose sharply once I settled into the field, after balancing the first two years against part-time work.
- Relevant mathematics coursework: Analysis I & II, Linear Algebra, Probability & Statistics.
- Relevant AI/ML coursework: Artificial Intelligence I, Artificial Intelligence II (Deep Learning for Natural Language Processing), Algorithms and Foundations of Machine Learning, Pattern Recognition (Machine Learning), Computational Machine Learning, Software Development for Information Systems.
- The degree includes the coursework required for a teaching license recognized by the Greek Ministry of Education, qualifying me to teach in schools.

Athens College (Hellenic-American Educational Foundation)

2022

High School Diploma, Grade: 19.8/20 (*“Excellent,” with Honors* / *“Μετ’ Επαίνου”*)

Research Experience

DIT AI Team, NKUA | Pre-graduate Research Associate

2024 – Present

Advisors: Prof. Manolis Koubarakis and Sergios Kefalidis (PhD candidate)

- Working on my undergraduate thesis, which develops a method for generating synthetic datasets to train and evaluate Knowledge Graph Question Answering (KGQA) systems. *Title: A Configurable Framework For Generating Synthetic KGQA Benchmarks*
- Assisting Sergios Kefalidis on his PhD research into KGQA systems, mainly with experiments and evaluation.

Research in Preparation

- Work on synthetic data generation for Knowledge Graph Question Answering, currently being written up with the DIT AI Team (NKUA).
- Work on an Optimizer, currently being developed, based on the paper "The Information Geometry of Local Generalization Dynamics" by Emmanouil M. Athanasakos (NKUA).

Conference Presentations

Co-authored the following peer-reviewed conference presentations in public mental health, where I handled the statistical analysis and the work on digital systems.

- Skali T., Mouliou M., Mavromichali K., **Spyrou G.**, Katsafadou V.S. “Evaluating the Impact of an Arts on Prescription Programme on Psychiatric Patients’ Well-Being.” *18th European Public Health Conference (EUPHA)*, Helsinki, Finland, 2025. Oral presentation (presented by T. Skali).
- Skali T., Theodoroulakis M., Sapouna V., **Spyrou G.** “The 10306 Psychosocial Support Helpline: A Scalable Model for Crisis Response and Brief Psychotherapy.” *18th European Public Health Conference (EUPHA)*, Helsinki, Finland, 2025. Oral presentation (presented by T. Skali).
- Sapouna V., **Spyrou G.** “The 10306 Psychosocial Support Helpline: The Digital Environment of the Service and the Use of New Technologies and Automated Processes for the Mental Health of Callers

and Staff.” 33rd Panhellenic Conference of the Hellenic Psychiatric Association, Greece. Chaired by M. Theodoroulakis.

Teaching Experience

Teaching Assistant (Lab), NKUA 2024 – 2026

- Delivered a guest lecture, "Debugging Memory Errors with Valgrind", to the full first-year cohort of the "Introduction to Programming" course (Dec. 2025).
- "Introduction to Programming" lab (Oct. 2025–Feb. 2026).
- "Data Structures and Programming Techniques" lab (Mar.–Jun. 2025).
- "Introduction to Programming" lab (Oct. 2024–Feb. 2025).
- "Data Structures and Programming Techniques" lab (Mar.–Jun. 2024).

Program Assistant, CTY Greece at Anatolia College Summer 2026

The Magic of Mathematical Thinking, Summer Program (Athens)

- Taught part of the curriculum myself and assisted the lead instructor with the rest, while also running the program’s daily activities.

Instructor, CTY Greece at Anatolia College Mar. 2026

Video Game Development, Weekend Program (Athens)

Teaching Assistant, CTY Greece at Anatolia College Apr. 2024

International Relations, Weekend Program (Athens)

Professional Experience

10306 Psychosocial Support Helpline 2023 – Present

Administrative & Technical Support Specialist (full-time, paid, held alongside full-time studies)

- Maintain and provide technical support for the 10306 Educational Toolkit and the service’s digital platform.
- Run the helpline’s social media and public outreach, and handle day-to-day technical work for the Greek Ministry of Health’s national service.

CTY Greece at Anatolia College | Residential Assistant Summers 2023, 2024

Older Students Residential Summer Program (Thessaloniki)

Technical Skills

- **Programming:** C, C++, Python (advanced); SQL, SPARQL, MATLAB (intermediate); Java (basic).
- **ML & Tools:** PyTorch, Large Language Models, Kaggle, Google Colab, LaTeX.

Languages

Greek: Native. **English:** Cambridge C2 Proficiency (2019). **German:** Goethe-Zertifikat B2 (2021).

Awards & Distinctions

- CTY Greece talent search recognition (SCAT & STB), 2017–2020.
- Cryptology summer studies, Athens, 2020; International Relations studies, Anatolia College, 2018–2019.
- Honorable Mention, “Kangaroo” Mathematics Competition (2016) and Panhellenic “Aristotelis” Physics Competition (2016).

Interests

- **Music:** Part-time Trombone studies (Philippos Nakas Conservatory), music theory, harmony and solfège.
- **Sports:** Football (A.E. Kerameikos, Athens FA); formerly also played tennis.